Spatio-Temporal Reconnection for Multi-Robot Networks Using Novel Prescribed-Time CBFs

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***Abstract*— In multi-robot systems, maintaining *persistent* communication graph connectivity is often overly restrictive, especially when robots have limited communication ranges but operate in large environments. Instead, allowing robots to temporarily disconnect and later reconnect is often more desirable for efficient task execution while still ensuring timely information sharing across the team. In this paper, we propose a novel prescribed-time control barrier function framework that enables robots to temporarily disconnect and re-enter the communication range within an adjustable and feasible prescribed time. Moreover, we introduce a reconnection triggering mechanism that jointly considers task execution and reconnection urgency, thereby providing a principled way to decide when reconnection should occur. Theoretical analysis justifies convergence to the satisfying reconnection within a prescribed finite time. Experimental results validate the performance of our proposed novel PT-CBF with improved task efficiency and satisfying reconnection.**

I. Background

We consider a team of N mobile robots with dynamics $\dot{\mathbf{x}}_i = f_i(\mathbf{x}_i) + g_i(\mathbf{x}_i)\mathbf{u}_i$, where $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{u}_i \in \mathbb{R}^q$ are the state and control input of robot i. In this paper, we adopt the distance-based communication model. Specifically, for each pair of robots (i, j), we define $h_{i,j}(\mathbf{x}) = R^2 - \|\mathbf{x}_i - \mathbf{x}_j\|^2$, where $R > 0$ is the communication range. Thus, $h_{i,j}(\mathbf{x}) \geq 0$ indicates that robots i and j are within communication range.

Problem Statement: In this paper, instead of guaranteeing the *global connectivity persistently* for the robot team to transmit information, we consider maintaining *joint global connectivity* for the multi-robot system, which allows robots to temporarily disconnect and later reconnect but still enables information to be synchronized periodically. Our goal is to determine *when and which* edge to connect using spatial and temporal information, while maintaining the whole communication graph jointly globally connected without overly disrupting task execution.

II. Method

A. Novel Prescribed-Time CBF:

To guarantee reconnection and adjust the prescribed time, we propose the following novel prescribed-time CBF constraint:

$$\dot{h}_{i,j}(\mathbf{x}) + c\,\mu(t)\,\mathrm{sign}(h_{i,j}(\mathbf{x}))|h_{i,j}(\mathbf{x})|^{\rho} \geq 0, \quad \rho \in (0, 1), \tag{1}$$

where

$$\mu(t) = \left(\frac{T_{\mathrm{p}}}{T_{\mathrm{p}} - t}\right)^m + \alpha, \qquad T_{\mathrm{p}} = \frac{|h_{i,j}(\mathbf{x}_0)|^{1-\rho}}{c(1-\rho)\mu_0}. \tag{2}$$

Here, $c > 0$ is a design parameter, $\rho \in (0, 1)$ sets the prescribed-time convergence rate, and $\mu(t)$ is a time-varying function that grows unbounded as $t \to T_{\mathrm{p}}$, with $m > 0$ and $\alpha > 0$ shaping its growth. Moreover, $\mathbf{x}_0$ is the state at the triggering instant and $\mu_0 = \mu(0)$. Thus, T_{p} is determined automatically from the current state, and a disconnected robot pair is guaranteed to re-enter the communication range R within time T_{p}.

B. Spatio-temporal Reconnection Mechanism:

To determine when and which edge to connect, we define a spatio-temporal edge weight

$$w_{i,j}(t) = \Gamma_{i,j}(\mathbf{x}, \mathbf{u}^{\mathrm{nom}}, t), \tag{3}$$

where $\Gamma_{i,j}(\cdot)$ captures both spatial reconnection difficulty and temporal urgency. Edge (i, j) is activated when $w_{i,j}(t) > 0$. Based on the activated edges, the final control input is obtained by solving the following Quadratic Programming (QP) problem:

$$\begin{aligned} \mathbf{u}^*(t) = \arg\min_{\mathbf{u}} \quad & \sum_{i=1}^{N} \|\mathbf{u}_i - \mathbf{u}_i^{\mathrm{nom}}(t)\|^2 \\ \text{s.t.} \quad & \text{safety control constraints,} \\ & \text{active connectivity control constraints} \end{aligned} \tag{4}$$

In this way, all desired edges are realized at least once per window, which enforces joint global connectivity.

III. Results and Discussion

Fig. 1 illustrates the performance of the proposed novel PT-CBF in multi-robot scenarios. The results suggest that it supports periodic information exchange without imposing overly restrictive persistent connectivity requirements. By allowing temporary disconnections and enforcing reconnection only when needed, the method offers greater flexibility in large environments with limited communication ranges.

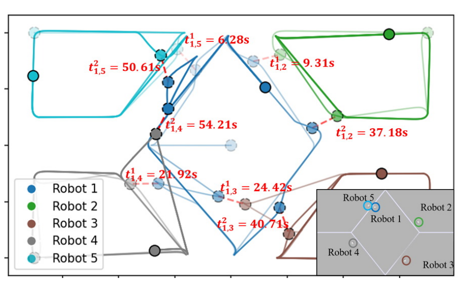

Fig. 1. Our Novel PT-CBF

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